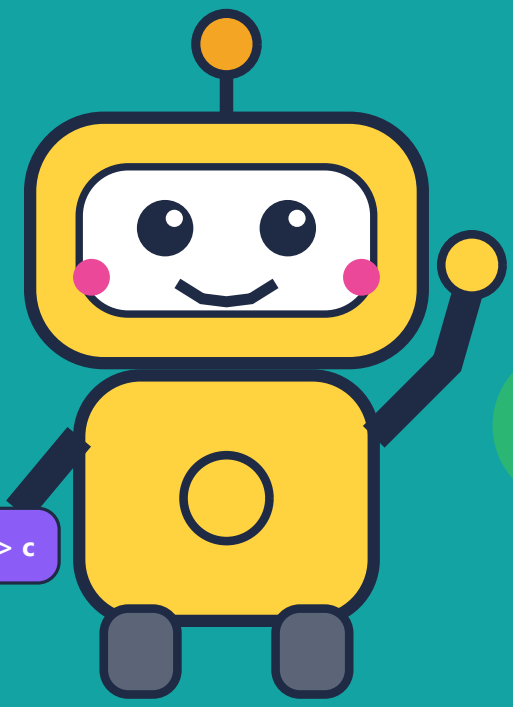
CodeSchool · Class 7

Think Like a Coder

Computational thinking and AI for Class 7

Learn it · See it · Try it · Build it



`sticks = 3n + 1` `Repeat until n is 0` `If a + b > c`

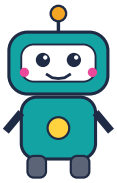
Follows NCERT Class 7 maths, CBSE's four AI chapters and two projects

Built for CBSE's Computational Thinking and AI curriculum, 2026-27

Theory, worked examples, practice, coding on paper, AI and projects

SAMPLE EDITION · DRAFT

© 2026 Bullseye Knowledge System Limited. All rights reserved.
CodeSchool is a working name. Not an official CBSE or NCERT publication.



Welcome back! I am Robo. In Class 7 we count in binary like a computer, write formulas with variables, solve logic puzzles by filtering cases, find out how AI predicts and why it can be unfair, and run two real projects.

How this book works

Learn	Robo explains one big idea in simple words, with the new coder words to remember.
Example	A worked example shows the idea step by step.
Try it!	Five practice questions. Circle the answer or write it in the box.
Now you build it	Coding on paper: put code cards in order, fix bugs and write your own programs.
Challenge	A harder puzzle for when you are ready for more.
How did I do?	Colour a face to show how sure you feel.
Projects	Two 20-hour team projects: plan, collect real data, build, test, present and score your work with a rubric.

My code cards

Coders use cards like these. Each colour means something different. You will find cut-out cards at the back of the book.

Step: do this **Repeat: a loop** **If: a decision** **Define: a function**

When: an event

For teachers and parents

Chapters 1-8 go with the same chapters of the NCERT Class 7 maths textbook, so they can be used in maths periods. AI chapters 1-4 follow the AI part of CBSE's Class 7 handbook, and the two projects run across the year. CBSE sets 100 hours a year in Class 7: about 40 for computational thinking, 20 for AI and 40 for two projects. The coding here needs no computer: students write, trace and debug programs on paper, which prepares them for block and text coding. Answers are at the back. Encourage children to explain their thinking, not just the answer.

© 2026 Bullseye Knowledge System Limited

What's inside

1. Large Numbers Around Us 4
2. Arithmetic Expressions 7
3. A Peek Beyond the Point 11
4. Expressions using Letter-Numbers 14
5. Parallel and Intersecting Lines 17
6. Number Play 20
7. A Tale of Three Intersecting Lines 23
8. Working with Fractions 26
AI 1. AI Domains and Applications 29
AI 2. AI in Industries 32
AI 3. Data Visualisation and Analysis 35
AI 4. Ethics and AI Bias Awareness 38
Project 1. Nutri-Quest Food Planner 41
Project 2. Sort It with AI 45
Answers 48
Coder words 50
Cut-out code cards 52

Coder skills in this book

Levels: C4 big numbers, expressions, binary, variables and fractions; C5 logic, angle and triangle rules, AI, data and projects.

Ch	Coder skill	Level	Ch	Coder skill	Level
1	Compare, round and estimate big numbers step by step	C4	8	Fraction operations and working backwards	C4
2	Order of operations and expression trees	C4	AI1	Classify, predict a number, or find groups	C5
3	Decimals, binary numbers and punched-card sorting	C4	AI2	AI in health, education, transport and communication	C5
4	Variables and formulas for growing patterns	C4	AI3	Choose bar, line or pie charts and read them honestly	C5
5	Angle rules and tracing drawing programs	C5	AI4	Spot bias in data and keep humans in charge	C5
6	True or false: list every case and filter with clues	C5	P1	Use data and rules to plan a balanced week	C5
7	Triangle rules as checks; debugging a function	C5	P2	Build a waste classifier and measure its accuracy	C5

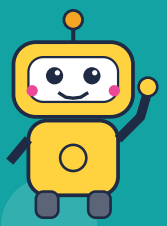
© 2026 Bullseye Knowledge System Limited

CHAPTER 1 · LEVEL C4

1 Large Numbers Around Us

Coder skill: Compare, round and estimate big numbers step by step

Goes with NCERT maths, Class 7, chapter 1

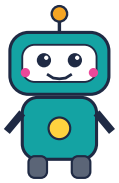


Learn

In the **Indian system** we group digits as lakhs and crores: 1 lakh = 1,00,000 and 1 crore = 1,00,00,000 = 100 lakh. In the **international system** we use thousands, millions and billions: 1 million = 10 lakh and 1 billion = 100 crore.

Computers compare numbers with a fixed set of steps: the number with **more digits** is bigger; if they have the same number of digits, compare **from the left** until two digits differ. To **estimate**, round to a convenient place first.

lakh	crore	million	estimate
1,00,000	1,00,00,000	10,00,000 (10 lakh)	a close, easy answer



To compare two big numbers I follow an algorithm: count the digits first, then compare from the left until two digits differ.

Example

Which is bigger: 4,56,78,912 or 45,678,921?

1. Remove the commas: 45678912 and 45678921. Both have **8 digits**, so we compare from the left.
2. 4 = 4, 5 = 5, 6 = 6, 7 = 7, 8 = 8, 9 = 9. The first difference is in the tens place: 1 against 2.
3. So **45,678,921** is bigger. Commas only help us read; Indian and international grouping show the same value.

★ *Coder fact!* Indian newspapers often give big numbers in lakhs and crores, while international ones use millions and billions.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 How many lakhs make 1 crore?

2 1 million is the same as ...

a) 1 lakh **b)** 10 lakh **c)** 1 crore **d)** 10 crore

3 Using the digits 7, 0 and 3, each exactly twice, write the largest and the smallest 6-digit numbers.

4 Round 4,86,72,315 to the nearest lakh.

5 What is the product of all the digits on a phone keypad, 0 to 9?

a) 362880 **b)** 3628800 **c)** 0 **d)** 45

Now you build it

An algorithm that compares two big numbers.

Input numbers A and B | **If one has more digits** | **say that one is bigger**

Repeat for each place from the left | **If the digits differ, the bigger digit wins; stop**

Trace it for 9,87,654 and 10,02,003. Then for 45,67,890 and 45,69,001. In which place is the first difference?

© 2026 Bullseye Knowledge System Limited

★ Challenge

India's population is about 140 crore. Write this in digits, and in the international system.

✓ How did I do?

I can read lakhs, crores and millions.	I can compare big numbers step by step.	I can round and estimate.
		

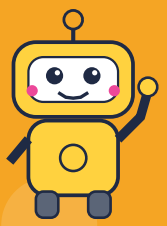
© 2026 Bullseye Knowledge System Limited

CHAPTER 2 · LEVEL C4

2 Arithmetic Expressions

Coder skill: Order of operations and expression trees

Goes with NCERT maths, Class 7, chapter 2

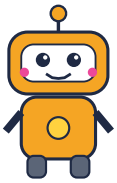


Learn

An **expression** is a mix of numbers and operations, like 8 + 6 x (5 - 2). To get one answer everyone must use the same **order**: **brackets** first, then **x and ÷** from left to right, then **+ and -** from left to right.

Computers store an expression as a **tree**: each operation joins two parts. They work it out from the bottom of the tree up, so the operation at the top is done last.

expression numbers joined by operations	**term** a part added or subtracted	**brackets** do this part first	**precedence** which operation goes first



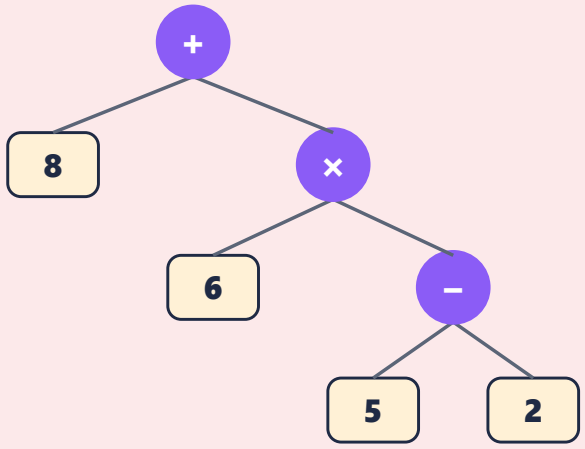
A computer reads 8 + 6 x 3 as 8 + 18, not 14 x 3. Brackets first, then x and ÷, then + and -. I draw it as a tree!

© 2026 Bullseye Knowledge System Limited

Example

Work out 8 + 6 x (5 - 2)

1. Brackets first: 5 - 2 = 3.
2. Then multiply: 6 x 3 = 18.
3. Then add: 8 + 18 = **26**. In the tree, the bottom operation (-) is done first and the top one (+) last.



★ *Coder fact!* Spreadsheets and scientific calculators follow the same order of operations you do: brackets first, then x and ÷, then + and -.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 Work out 20 - 4 x 3 and (20 - 4) x 3.

2 Which operation in the box makes this true? 6 [?] 4 + 10 = 34

a) + **b)** - **c)** x **d)** ÷

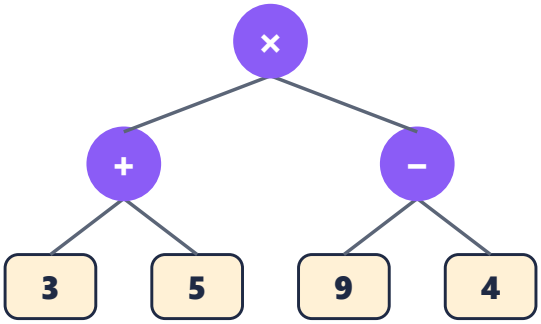
3 A bus has 40 seats and 23 passengers. At stop 1, 5 get off and 8 get on. At stop 2, 6 get off and 12 get on. How many seats are empty now?

4 Where do brackets go to make 2 + 3 x 4 equal 20?

a) (2 + 3) x 4 **b)** 2 + (3 x 4)

c) no brackets are needed

5 What is the value of this expression tree?



© 2026 Bullseye Knowledge System Limited

Now you build it

A program that adds up a shopping bill.

- Set total to 0
- change total by 3 x 45
- change total by 2 x 120
- change total by -50 (discount)
- say total

What does it say? Write the whole bill as one expression.

Challenge

Use 2, 3, 4 and 5, each once, with any operations and brackets, to make 24.

How did I do?

I can follow the order of operations.	I can use brackets to change the order.	I can read an expression tree.
		

© 2026 Bullseye Knowledge System Limited

CHAPTER 3 · LEVEL C4

3 A Peek Beyond the Point

Coder skill: Decimals, binary numbers and punched-card sorting

Goes with NCERT maths, Class 7, chapter 3



Learn

In our **decimal** system each place is worth 10 times the place to its right: ..., 100, 10, 1, and after the point 1/10, 1/100, ... So 3.47 means 3 ones, 4 tenths and 7 hundredths.

Computers use the **binary** system, where place values are 1, 2, 4, 8, 16, ... Each **bit** is 0 or 1. 23 = 16 + 4 + 2 + 1, so 23 is **10111** in binary. Five bits give 32 patterns, enough to give each of the 26 letters its own punched card.

decimal place values in tens	**binary** place values in twos	**bit** a binary digit, 0 or 1	**place value** what a digit's position is worth



Our numbers use bundles of ten. Computers use bundles of two: 1, 2, 4, 8, 16. That is binary, written with only 0 and 1!

Example

Writing 23 in binary with dot cards

1. Start from the biggest card that fits: 16. That leaves 7.
2. 8 is too big, so the 8 card is face down: 0. Then 4 (leaves 3), 2 (leaves 1) and 1.
3. Face-up cards are 1s and face-down cards are 0s: 23 = **10111**.



★ *Coder fact!* Acharya Pingala used patterns of short and long syllables, much like binary, to study poetry metres about 2,000 years ago.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 What is 13 in binary?

a) 1011 **b)** 1101 **c)** 1110 **d)** 1001

2 What is the binary number 10010 in decimal?

3 Every student in a class of 40 needs a different binary number. How many bits are needed?

a) 5 **b)** 6 **c)** 7 **d)** 8

4 Which is the smallest?

a) 0.5 **b)** 0.45 **c)** 0.405

5 Find the next term: 1.2, 2.4, 2.6, 5.2, 5.4, 10.8, ?

a) 11.0 **b)** 21.6 **c)** 10.10 **d)** 11.8

Now you build it

An algorithm that turns a decimal number into binary by halving.

Input n **Repeat until n is 0** **write down the remainder of n ÷ 2**

set n to n ÷ 2, dropping the remainder **read the remainders from last to first**

Trace it for 14. Write each value of n and each remainder. What binary number do you get?

Challenge

Letters are numbered A = 1 to Z = 26 and written in 5-bit binary (J = 10 = 01010). Which letter is 10011? Write the 5 bits for R.

© 2026 Bullseye Knowledge System Limited

✓ How did I do?

I can read decimal place values.	I can change numbers between binary and decimal.	I can explain how 5 bits can label 26 letters.
		

© 2026 Bullseye Knowledge System Limited

CHAPTER 4 · LEVEL C4

4 Expressions using Letter-Numbers

Coder skill: Variables and formulas for growing patterns

Goes with NCERT maths, Class 7, chapter 4

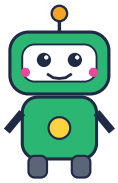


Learn

A **letter-number** (a variable) stands for a number. A **formula** uses letters to give a rule that works every time: the perimeter of a square with side s is **4s**.

To use a formula, **substitute**: put the number in place of the letter. Programs do the same: *set sticks to 3 x n + 1* works for any n. Terms with the same letter are **like terms** and can be added: 3x + 2x = 5x.

variable	formula	substitute	like terms
a letter that stands for a number	a rule written with letters	put a number in place of a letter	terms with the same letter



A letter-number is a variable: a name for a number that can change. Write the rule once, then put in any number!

Example

Matchstick squares in a row

1. 1 square needs 4 sticks, 2 squares need 7 and 3 squares need 10. Each new square adds **3** sticks.
2. So n squares need **3n + 1** sticks: 3 for each square, plus 1 for the very first side.
3. Substitute n = 20: 3 x 20 + 1 = **61** sticks, without drawing them.



square: 4 sticks 2 squares: 7 sticks 3 squares: 10 sticks

★ *Coder fact!* In games, variables like score and lives are letter-numbers: names for values that change as you play.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 If a = 4 and b = 3, what is 5a - 2b?

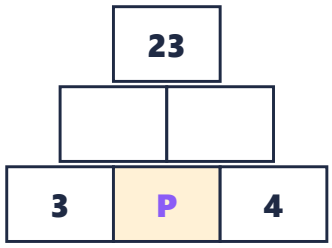
2 Simplify 3x + 5 + 2x - 1.

a) 5x + 4 **b)** 6x + 4 **c)** 5x + 6 **d)** 10x

3 A + B = Z, where B = 3 x A, B is more than 6 and Z is less than 13. All are whole numbers. What is A?

a) 1 **b)** 2 **c)** 3 **d)** 4

4 Each block is the sum of the two blocks below it. What is P?



5 A pen costs Rs p and a notebook costs 3 times as much. Riya buys 2 pens and 4 notebooks. Which expression gives the total?

a) 14p **b)** 6p **c)** 9p **d)** 2p + 4

Now you build it

A program that uses the matchstick formula.

Input n (number of squares) | set sticks to 3 x n + 1 | If sticks > 50

say 'Buy another box' | say sticks

Run it for n = 10 and for n = 17. What does it say each time?

© 2026 Bullseye Knowledge System Limited

★ Challenge

Triangles in a row: 1 triangle needs 3 sticks, 2 need 5, 3 need 7. Write the formula. How many sticks for 25 triangles?

✓ How did I do?

I can use letters for numbers.	I can substitute values into a formula.	I can write a formula for a growing pattern.
		

© 2026 Bullseye Knowledge System Limited

CHAPTER 5 · LEVEL C5

5 Parallel and Intersecting Lines

Coder skill: Angle rules and tracing drawing programs

Goes with NCERT maths, Class 7, chapter 5

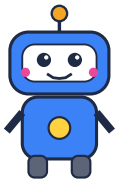


Learn

Parallel lines never meet. When two lines **intersect**, **vertically opposite** angles are equal, and angles next to each other on a straight line add up to **180°**.

A line that crosses two parallel lines is a **transversal**: it makes 8 angles, and they come in just two sizes that add up to 180°. In a drawing program, lines drawn with the same heading are parallel, so you can check a program by tracing the heading after each turn.

parallel never meet	**intersect** cross at a point	**vertically opposite** equal angles across a crossing	**transversal** a line crossing two lines

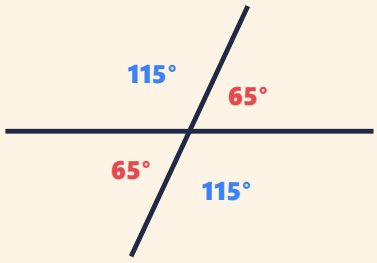


Two lines drawn with the same heading never meet: they are parallel. When lines cross, I use angle rules instead of measuring!

Example

Two lines cross, and one angle is 65°

1. The angle next to it on the straight line is 180° - 65° = **115°**.
2. Vertically opposite angles are equal, so the other two angles are **65°** and **115°**.
3. Check: 65 + 115 + 65 + 115 = 360°, a full turn.



★ *Coder fact!* Railway lines are parallel: the two rails must stay exactly the same distance apart.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 Two angles on a straight line: one is 72°. What is the other?

2 Vertically opposite angles are (3x + 10)° and (5x - 30)°. Find x and the size of each angle.

3 At which times is the minute hand on 12 and the hour hand perpendicular to it?

a) 3:00 and 9:00 **b)** 6:00 only **c)** 12:00 and 6:00 **d)** 3:00 only

4 A turtle program: *forward 50, turn right 90°, forward 30, turn right 90°, forward 50.* How are the first and last lines related?

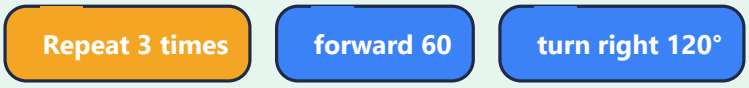
a) parallel **b)** perpendicular **c)** they meet

5 Each side of a square is divided into 4 equal parts, and lines parallel to the sides join the marks. How many squares of all sizes are there?

a) 16 **b)** 25 **c)** 30 **d)** 32

Now you build it

Trace this drawing program on dotted paper.



What shape does it draw? Then write a program for a rhombus with sides of 60 whose opposite sides are parallel.

© 2026 Bullseye Knowledge System Limited

★ Challenge

A transversal crosses two parallel lines. One of the 8 angles is 50°. How many angles are 50°, and what are the others?

✓ How did I do?

I can use vertically opposite and straight-line angles.	I can spot parallel lines in a drawing program.	I can find angles made by a transversal.
		

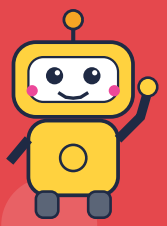
© 2026 Bullseye Knowledge System Limited

CHAPTER 6 · LEVEL C5

6 Number Play

Coder skill: True or false: list every case and filter with clues

Goes with NCERT maths, Class 7, chapter 6

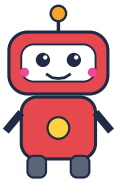


Learn

Each light is either ON or OFF: **true** or **false**. With 3 lights there are 2 x 2 x 2 = **8** possible cards; with 5 lights, 32. On a punched card a notch means ON and a hole means OFF.

To solve a logic puzzle, start with **every case**, then **filter**: each clue removes the cards that break it. A computer does exactly this, very fast. Some number facts are true for **every** case, and we can explain why, like 'a 2-digit number minus its reverse is always a multiple of 9'.

true or false the two values in logic	**case** one possible combination	**filter** remove cases that break a rule	**truth table** a list of every case



With 3 lights there are only 8 ON/OFF cards. Each clue throws away the cards that break it, until one card is left. That's how computers reason!

Example

Three lights R, G and B: which are ON?

1. Clue 1: *Red is ON.* Keep the 4 cards that start with U.
2. Clue 2: *Blue is OFF.* Keep UOO and UUO.
3. Clue 3: *Exactly two lights are ON.* Only **UUO** is left: Red and Green are ON, Blue is OFF.



★ *Coder fact!* Inside a computer, every decision comes down to tiny switches that are either on or off: true or false.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 How many ON/OFF combinations are there for 4 lights?

2 With the 8 cards for R, G and B, the clue is *Green is OFF*. How many cards are left?

a) 2 **b)** 4 **c)** 6 **d)** 8

3 Clues: *Red is ON* and *all lights are the same*. Which card is left?

a) all ON **b)** all OFF **c)** only Red ON

4 A 2-digit number with different digits minus its reverse is always a multiple of ...

a) 2 **b)** 5 **c)** 9 **d)** 10

5 A 4-digit number has two even and two odd digits. Can its digit sum be odd?

a) yes **b)** no, it is always even

Now you build it

A computer can solve a number puzzle by checking every case.

Repeat for every number from 10 to 99 **If its digits add up to 11**

If it is 27 more than its reverse **say the number**

Which numbers does the program say? Find them without checking all 90 numbers.

Challenge

With 5 lights (Y, P, R, G, B), how many cards are in the full deck? Clues: Y is ON, P is OFF, exactly 3 lights are ON. How many cards are left?

© 2026 Bullseye Knowledge System Limited

✓ How did I do?

I can list all true/false combinations.	I can use clues to filter cases.	I can explain why a rule is always true.
		

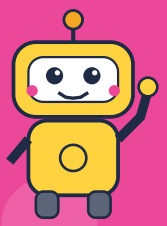
© 2026 Bullseye Knowledge System Limited

CHAPTER 7 · LEVEL C5

7 A Tale of Three Intersecting Lines

Coder skill: Triangle rules as checks; debugging a function

Goes with NCERT maths, Class 7, chapter 7

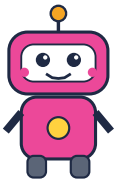


Learn

Three lengths make a triangle only if **each side is shorter than the other two added together** (the **triangle inequality**). The three angles of a triangle always add up to **180°**.

Triangles are named by sides (**equilateral**: 3 equal, **isosceles**: 2 equal, **scalene**: none equal) and by angles (acute, right, obtuse). A program that checks inputs must test **every** condition; testing only one is a common bug.

triangle inequality	equilateral	isosceles	scalene
each side < sum of the other two	all sides equal	two sides equal	no sides equal



Not any three sticks make a triangle! My is_triangle() check has to test all three sides, or it has a bug.

Example

Can these lengths make a triangle?

1. 3 cm, 4 cm, 8 cm: 3 + 4 = 7, which is less than 8, so the two short sides cannot meet. **No triangle.**
2. 5 cm, 6 cm, 8 cm: 5 + 6 = 11 > 8, 5 + 8 > 6 and 6 + 8 > 5. **A scalene triangle.**
3. Shortcut: if the two shortest sides add up to more than the longest, it works.

Coder fact! Bridges and towers use triangles because a triangle cannot change shape unless a side changes length.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 Which set of lengths can form a triangle?

a) 2, 3, 6 **b)** 4, 4, 9 **c)** 5, 7, 10 **d)** 1, 2, 3

2 Two angles of a triangle are 48° and 67°. What is the third?

3 A triangle has angles of 30°, 60° and 90°. What kind is it?

a) acute **b)** right-angled **c)** obtuse

4 An isosceles triangle has a perimeter of 20 cm, whole-number sides, and its two equal sides are even numbers. How many such triangles are there?

a) 1 **b)** 2 **c)** 3 **d)** 4

5 An equilateral triangle and a square both have sides of 4 cm. Which has the bigger area?

a) the square **b)** the triangle **c)** the same

Now you build it

This function is meant to check whether three lengths make a triangle.

Define is_triangle(a, b, c) If a + b > c give back 'yes' else give back 'no'

Test it with 2, 10, 3. What does it give back, and what is the right answer? Fix the bug.

Challenge

How many different triangles with whole-number sides have a perimeter of 12 cm?

© 2026 Bullseye Knowledge System Limited

✓ How did I do?

I can test the triangle inequality.	I can use the angle sum of 180°.	I can debug a checking function.
		

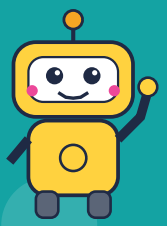
© 2026 Bullseye Knowledge System Limited

CHAPTER 8 · LEVEL C4

8 Working with Fractions

Coder skill: Fraction operations and working backwards

Goes with NCERT maths, Class 7, chapter 8

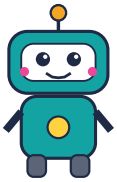


Learn

'Of' means multiply: 3/4 of 48 = 3/4 x 48 = 36. To multiply fractions, multiply tops and bottoms. To divide by a fraction, multiply by its **reciprocal**: 6 ÷ 3/4 = 6 x 4/3 = 8.

Many puzzles give the end and ask for the start. **Work backwards**: undo each step with its **inverse** (add undoes subtract, ÷ undoes x), starting from the last step.

of means multiply	**reciprocal** the fraction turned upside down	**inverse** the step that undoes another	**work backwards** start from the end



To undo a program, I do the opposite steps in the opposite order. 'Take away a third' is undone by 'divide by 2/3'!

Example

Meera's sweets

1. Meera keeps 1/3 of her sweets, gives 1/4 of the rest to her sister, and gives the remaining 18 to her brother. How many did she have?
2. The brother got 3/4 of the rest, so the rest was 18 ÷ 3/4 = **24**.
3. The rest was 2/3 of the total, so the total was 24 ÷ 2/3 = **36**. Check: she keeps 12, her sister gets 6 and her brother 18.

★ *Coder fact!* Recipes are scaled with fractions: half a recipe means half of every ingredient.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 What is 3/4 of 48?

2 Work out 2/3 x 9/10.

a) 3/5 **b)** 11/13 **c)** 18/13 **d)** 2/5

3 On a bus, 1/4 of the passengers get off at stop A, then 20 get on. At stop B, half get off. 40 are left. How many were on the bus at the start?

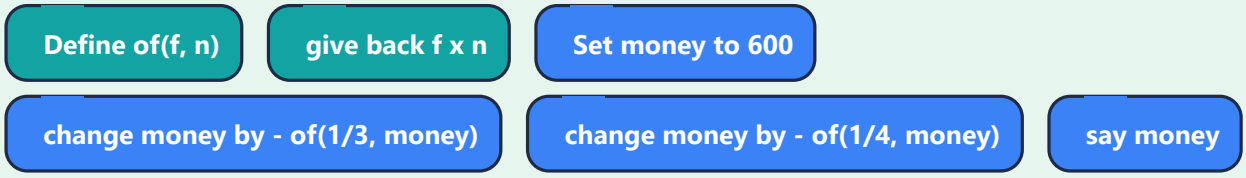
4 Work out 6 ÷ 3/4.

a) 8 **b)** 4.5 **c)** 2 **d)** 18

5 Half of a field grows wheat. 2/5 of the wheat part is watered today. What fraction of the whole field is watered?

Now you build it

A fraction function used inside a program.



Trace it. What is money after each step, and what does the program say?

© 2026 Bullseye Knowledge System Limited

★ Challenge

Write the 'undo' program: start from 300 and get back to 600.

✓ How did I do?

I can find a fraction of a quantity.	I can multiply and divide fractions.	I can work backwards to find a start value.
		

© 2026 Bullseye Knowledge System Limited

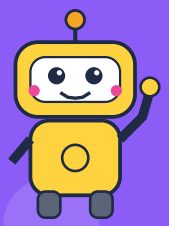
AI1

AI CHAPTER 1 · LEVEL C5

AI Domains and Applications

AI skill: Classify, predict a number, or find groups

Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 1



Learn

AI works in three main **domains**: **Data Science** (numbers and tables), **Computer Vision** (images and video) and **Natural Language Processing** (text and speech). Each learns patterns from a **dataset**.

Classification puts things into labelled groups (spam or not spam). **Regression** predicts a number (tomorrow's temperature). **Clustering** finds groups in data that has no labels. A dataset is split into **training**, **validation** and **test** parts, so the model is checked on data it has never seen.

classification sort into labelled groups	**regression** predict a number	**clustering** find groups without labels	**dataset** the data a model learns from



AI has three big areas: numbers and tables, pictures, and language. And three ways to predict: classify, regress and cluster!

Example

Predicting a number from a small dataset (regression)

1. Hours of practice and test marks for one student: 1 hour: 52, 2 hours: 58, 3 hours: 64, 4 hours: 70.
2. The pattern: each extra hour adds about **6** marks.
3. Prediction for 5 hours: about 70 + 6 = **76**. It is a prediction, not a promise: real data is rarely this neat.

Hours	Marks	Change
1	52	
2	58	+6
3	64	+6
4	70	+6
5	?	predict 76

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 Which technique predicts a number, like tomorrow's temperature?

a) classification **b)** regression **c)** clustering

2 A photo app groups pictures of the same face without being told any names. This is ...

a) classification **b)** regression **c)** clustering

3 An app reads car number plates from camera pictures. Which domain is this?

a) Data Science **b)** Computer Vision **c)** NLP

4 500 images are split into 70% training, 15% validation and 15% test. How many test images are there?

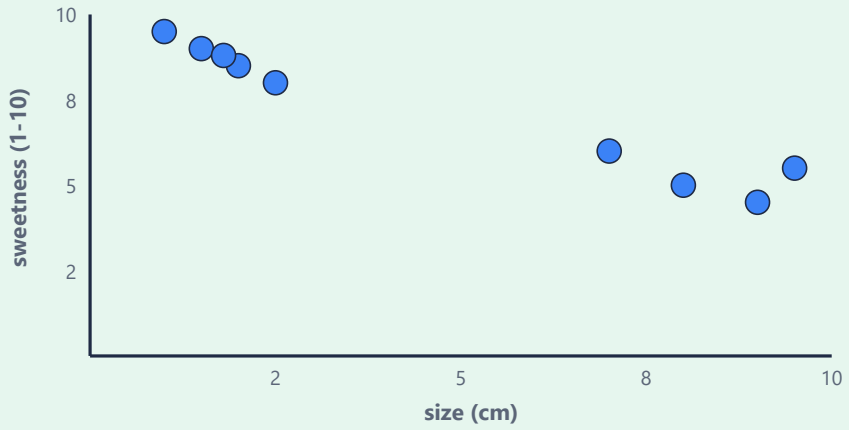
5 A chatbot answers questions typed in Hindi. Which domain is this?

a) Data Science **b)** Computer Vision **c)** NLP

© 2026 Bullseye Knowledge System Limited

Now you build it

Be a clustering algorithm. Each dot is a fruit, placed by its size and sweetness. There are no labels.



Look for points that sit close together

Repeat until every point is in a group

put the point with its nearest group

Name each group by what its points share

Circle the groups you find. How many are there, and what could each group be?

Challenge

Give one example from your school of classification, one of regression and one of clustering.

How did I do?

I can name three AI domains.	I can tell classification, regression and clustering apart.	I can explain training, validation and test data.
☺ 😐 ☹	☺ 😐 ☹	☺ 😐 ☹

© 2026 Bullseye Knowledge System Limited

AI2

AI CHAPTER 2 · LEVEL C5

AI in Industries

AI skill: AI in health, education, transport and communication

Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 2

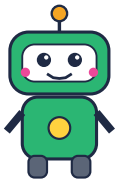


Learn

In **healthcare**, AI helps read X-rays and scans and spot illness early. In **education**, it gives personalised practice and instant feedback. In **transport**, it times traffic lights and finds faster routes. In **communication**, it translates, cleans up noisy audio and blocks spam.

AI is useful because it handles huge amounts of data quickly. But it can make mistakes, so people stay **responsible**: a doctor checks a diagnosis, and personal data must be kept safe.

diagnosis finding out what illness it is	**personalised** made for one person	**route planning** choosing the best way	**translation** changing one language to another



AI helps doctors, teachers, drivers and translators. But it helps people decide; it doesn't take the blame or the responsibility!

Example

How a route app chooses

1. It collects live data: how fast phones on each road are moving.
2. It predicts the travel time for each route, not just the distance.
3. It picks the shortest **time**, Route B here, and re-plans if there is an accident.

Route	Distance	Predicted time
A	12 km	35 min
B	15 km	28 min
C	10 km	40 min

★ *Coder fact!* Navigation apps use live data from many phones to estimate traffic and travel times.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 Analysing MRI scans helps which field?

a) healthcare **b)** education **c)** transport **d)** communication

2 Giving instant feedback on a quiz is AI in ...

a) healthcare **b)** education **c)** transport **d)** communication

3 Changing traffic-light timing using camera counts of cars is AI in ...

a) healthcare **b)** education **c)** transport **d)** communication

4 Using the route table in the example, which route would the app choose if it wanted the shortest **distance**?

a) A **b)** B **c)** C

5 Why should a doctor check an AI's diagnosis?

a) AI can make mistakes and the doctor is responsible

b) AI is always wrong

c) to make it slower

Now you build it

Rules for a smart traffic light at a crossing of two roads.

Count the cars waiting on each road

If one road has more than 20 cars waiting

give that road a longer green light

else

use the normal timing

If an ambulance is detected

turn its road green at once

Road 1 has 25 cars and road 2 has 8. What happens? Then an ambulance arrives on road 2. What happens now?

© 2026 Bullseye Knowledge System Limited

★ Challenge

Which check should come first in the program: the ambulance or the car count? Why?

✓ How did I do?

I can give examples of AI in four industries.	I can explain how a route app decides.	I can say why humans stay responsible.
		

© 2026 Bullseye Knowledge System Limited

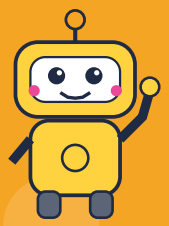
AI3

AI CHAPTER 3 · LEVEL C5

Data Visualisation and Analysis

AI skill: Choose bar, line or pie charts and read them honestly

Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 3

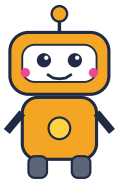


Learn

Structured data fits in rows and columns, like a marks table; **unstructured data** does not, like photos or voice messages. We collect data by observation, surveys, measurement and digital devices, and store it in databases, spreadsheets or CSV files.

Charts make patterns easy to see: **bar charts** compare groups, **line graphs** show change over time, and **pie charts** show parts of a whole (each slice's angle = its share of 360°). Good data is **accurate** (close to the true value) and **precise** (exact), and an honest chart does not exaggerate.

structured data	line graph	pie chart	accuracy
data in rows and columns	shows change over time	shows parts of a whole	closeness to the true value



A bar chart compares, a line graph shows change over time, and a pie chart shows parts of a whole. Pick the right one!

Example

How 40 students travel to school

1. Bus 20, walk 10, car 6, cycle 4. Total 40.
2. Each slice's angle = count ÷ 40 x 360°: bus 20 ÷ 40 x 360 = **180°**, walk 90°, car 54° and cycle 36°.
3. Check: 180 + 90 + 54 + 36 = 360°. Half the class comes by bus.



- **Bus: 20 (180°)**
- **Walk: 10 (90°)**
- **Car: 6 (54°)**
- **Cycle: 4 (36°)**

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 Which chart is best for showing a student's marks over six months?

a) pie chart **b)** line graph **c)** bar chart

2 Which chart is best for showing how a day's 24 hours are spent?

a) pie chart **b)** line graph **c)** scatter plot

3 In a pie chart of 60 students, 15 chose cricket. What angle is the cricket slice?

4 A scale always shows 50.12 kg for a bag that really weighs 48 kg. The scale is ...

a) precise but not accurate **b)** accurate but not precise

c) both **d)** neither

5 Which of these is structured data?

a) a table of marks **b)** a video

c) a voice message **d)** a photo

Now you build it

A shop's ice-cream sales from Monday to Sunday: 12, 15, 14, 20, 26, 40, 38.

Choose the chart type **Draw the axes and choose a scale** **Plot each value**

Join the points **Write the trend in one sentence**

Draw the chart in the space and write the trend.

© 2026 Bullseye Knowledge System Limited

★ Challenge

A bar chart's axis starts at 90 instead of 0, so a score of 92 looks twice as tall as 91. Why is this misleading, and how would you fix it?

✓ How did I do?

I can choose the right chart.	I can work out pie chart angles.	I can read a trend and check that a chart is honest.
		

© 2026 Bullseye Knowledge System Limited

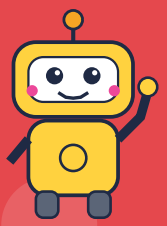
AI4

AI CHAPTER 4 · LEVEL C5

Ethics and AI Bias Awareness

AI skill: Spot bias in data and keep humans in charge

Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 4

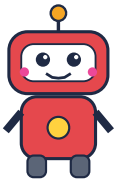


Learn

AI ethics means making sure AI is fair, open about how it decides, safe and respectful of privacy. **Bias** is an unfair preference for or against someone. Machines do not invent bias: it usually comes from the **data**.

Data bias: some groups are missing or rare in the training data. **Historical bias**: the data repeats past unfair decisions. **Measurement bias**: the wrong thing is measured. For important decisions, a **human in the loop** must check and take responsibility.

bias an unfair preference	**fairness** works equally well for everyone	**transparency** being open about how it decides	**human in the loop** a person checks the decision



I learn only from the examples people give me. If the examples are lopsided, my answers will be lopsided too!

Example

The lopsided deck

1. A classifier learns 'fruit or vegetable' from 18 fruit cards and only 2 vegetable cards.
2. It learns that almost everything is a fruit. On a test of 50 fruits and 10 vegetables it gets 45 fruits but only 3 vegetables right.
3. Overall it is 48 of 60 = 80% right, which looks good, but it is only **30%** right for vegetables. Fix: add vegetable examples and check each group.

Group	Training cards	Right in the test
Fruit	18	45 of 50 (90%)
Vegetable	2	3 of 10 (30%)

★ *Coder fact!* India's Digital Personal Data Protection Act, 2023 sets rules for how organisations may use people's personal data.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 A hiring system learnt from past decisions in which mostly men were hired, and now prefers men. This is ...

a) data bias
b) historical bias
c) measurement bias

2 A face recogniser trained mostly on one group of people often fails for others. This is ...

a) data bias
b) historical bias
c) measurement bias

3 A health AI judges how ill people are from how much money they spent on treatment. This is ...

a) data bias
b) historical bias
c) measurement bias

4 An AI rejects a loan application. What should happen next?

a) a person should review the decision
b) nothing: the AI is always right
c) the applicant should never apply again

5 Before sharing a shocking news post about a classmate, you should ...

a) check it with a trusted source, and not share hurtful posts
b) share it quickly
c) add your own comments

© 2026 Bullseye Knowledge System Limited

Now you build it

A fairness check you can run on any classifier.

Count each group in the training data

If one group is much smaller

add more examples of that group

Test on a balanced test set

Compare accuracy for each group

After adding 16 vegetable cards, the classifier gets 46 of 50 fruits and 9 of 10 vegetables right. Work out the accuracy for each group and overall. Is it fairer?

Challenge

Write two rules a school should follow before using AI to help choose students for a competition.

How did I do?

I can explain three kinds of bias.	I can check accuracy for each group.	I can say why humans must review important decisions.
		

© 2026 Bullseye Knowledge System Limited

P1

PROJECT 1 · 20 HOURS · LEVEL C5

Nutri-Quest Food Planner

Project skill: Use data and rules to plan a balanced week

CBSE: 40 project hours a year in Class 7, as two projects of 20 hours



The big question

Can we use nutrition data and simple rules to plan a balanced, affordable week of school lunches?

Your team collects data for 12 common foods: food group, protein, energy and cost per serving, from food packets or the Indian Food Composition Tables of ICMR-NIN, with your teacher's help.

Then write rules, for example: *every lunch has a grain, a protein food and a vegetable; no dish more than twice a week; each lunch costs at most Rs 40.* Plan Monday to Friday, check every day against every rule, fix what breaks, improve the plan and present it with charts.

nutrient	constraint	budget	balanced
what food gives the body	a rule a plan must follow	the most you can spend	has every food group

My project plan

Weeks	Stage	What I do	Done
1	**Define**	Agree the question and the food groups. Decide what data to collect for each food.	☐
2	**Collect**	Record food group, protein, energy and cost per serving for 12 foods. Write each source.	☐
3	**Organise**	Make one table of all the foods. Check for missing or odd values.	☐
4	**Write rules**	Write 3 to 5 rules as 'every lunch must ...' or 'if ... then ...'.	☐
5	**Build**	Make a first plan for Monday to Friday.	☐
6	**Check**	Check each day against each rule in a table. Fix every broken rule.	☐
7	**Improve**	Make version 2: cheaper, more protein or more variety. Compare the two versions.	☐
8	**Show**	Draw a bar chart of protein per day and a pie chart of cost by food group.	☐
9	**Present**	Present the rules, the plan, the checks and the charts in 3 minutes.	☐
10	**Reflect**	Score your work with the rubric and write what you would change.	☐

© 2026 Bullseye Knowledge System Limited

How a coder does it

Checking a plan against the rules (practice data)

1. Rules: R1 grain + protein + vegetable every day; R2 no dish more than twice a week; R3 at most Rs 40 a lunch.
2. Wednesday breaks R3 (Rs 46): swap paneer for chana (Rs 36). Friday breaks R2 (dal a third time): swap dal for chana, its second time.
3. Check again: every day now passes every rule. A computer checks plans the same way, rule by rule.

Day	Lunch	Rules broken
Mon	roti, dal, bhindi (Rs 32)	none
Tue	rice, rajma, salad (Rs 38)	none
Wed	roti, paneer, peas (Rs 46)	R3
Thu	rice, dal, cabbage (Rs 30)	none
Fri	roti, dal, spinach (Rs 30)	R2

Food data sheet

One row per food. Write values per serving and where each value came from.

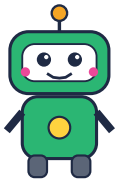
Food	Food group	Protein (g)	Energy (kcal)	Cost (Rs)	Source

© 2026 Bullseye Knowledge System Limited

My 5-day plan and rule check

Draw a table with a row for each day (Monday to Friday) and columns for the lunch, its cost and each rule. Tick each rule the day passes; mark any rule it breaks, fix the day and check again.

© 2026 Bullseye Knowledge System Limited



A planner is an algorithm with rules. Check every rule for every day, and fix the plan until nothing breaks!

★ How good is my project?

Score each row from 1 (just starting) to 4 (excellent). Then your teacher scores it too. Talk about any row where the scores differ.

What we check	What excellent (4) looks like	Me	Teacher
Question and rules	Clear rules that make a lunch balanced and affordable.	1 2 3 4	1 2 3 4
Data	Nutrition and cost data for every food, each with a source.	1 2 3 4	1 2 3 4
Plan	Every day follows every rule, and the checks are shown.	1 2 3 4	1 2 3 4
Improvement	Version 2 is better in a way that can be measured.	1 2 3 4	1 2 3 4
Charts and talk	The right chart for each job; a clear, short talk.	1 2 3 4	1 2 3 4
Teamwork	Everyone has a job and the work is shared fairly.	1 2 3 4	1 2 3 4

✓ Fair, safe and honest?

- [ ] We wrote down where every value came from.
- [ ] We respected food choices for religion, health and allergies.
- [ ] We did not collect anyone's weight or health details.
- [ ] We said our plan is a school project, not medical advice.

Looking back

Which rule was the hardest to keep, and why?

How would your plan change for a classmate with a food allergy?

© 2026 Bullseye Knowledge System Limited

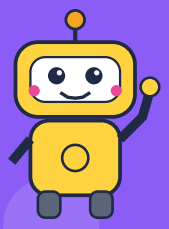
P2

PROJECT 2 · 20 HOURS · LEVEL C5

Sort It with AI

Project skill: Build a waste classifier and measure its accuracy

CBSE: 40 project hours a year in Class 7, as two projects of 20 hours



The big question

Can we teach a classifier to put school waste in the right bin, and measure how often it is right?

Waste is easier to recycle when it goes in the right bin: **wet** (food and garden waste), **dry** (paper, plastic, metal, glass) or **hazardous** (batteries, bulbs, medicines). Your team makes 60 picture cards of waste items and labels each one.

Use 40 cards to find rules (training) and keep 20 for testing. Write the rules as a decision tree, test it, measure the accuracy, study the mistakes, improve, and test again. Schools with computers can also train a no-code model on photos and compare.

label the right answer for an example	**feature** something you can observe	**accuracy** correct ÷ total	**test set** examples kept back for checking

My project plan

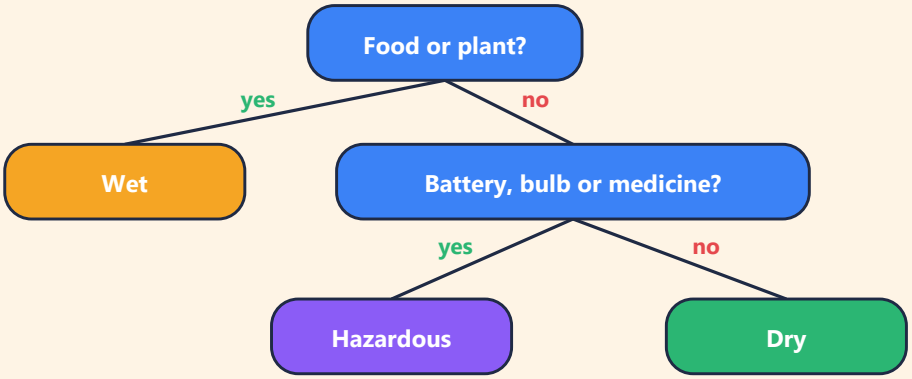
Weeks	Stage	What I do	Done
1	Define	Agree the question and the three bins. Decide how you will measure success.	☐
2	Collect	Make 60 picture cards of waste items, about 20 for each bin, and label each one.	☐
3	Check labels	Check unclear labels with your teacher. Shuffle and split: 40 training, 20 test.	☐
4	Find features	Study the training cards: what can you observe that separates the bins?	☐
5	Build	Write your rules as a decision tree with yes/no questions.	☐
6	Test	Run the tree on the 20 test cards. Record each prediction and the true label.	☐
7	Measure	Work out the accuracy. Make a table of which bins were mixed up.	☐
8	Improve	Fix a rule or add training cards, then test on new cards you have not used.	☐
9	Present	Show the tree, the results before and after, and what you learnt.	☐
10	Reflect	Score your work with the rubric. Which mistakes matter most?	☐

© 2026 Bullseye Knowledge System Limited

How a coder does it

Testing a decision tree (practice data)

1. The tree asks: *Is it food or a plant?* If yes: wet. If no: *Is it a battery, bulb or medicine?* If yes: hazardous; if no: dry.
2. On 20 test cards it got 17 right: accuracy = 17 ÷ 20 = **85%**.
3. The 3 mistakes show what to fix: a used tissue with food, a broken thermometer and a paint tin had no question that fitted them.



Test results

One row per test card. Write the true label, your tree's prediction, and whether it was right.

Card	Item	True bin	Predicted bin	Right? (Y/N)

© 2026 Bullseye Knowledge System Limited



A classifier is only as good as its examples. Test it on cards it has never seen, count the mistakes, and improve!

★ How good is my project?

Score each row from 1 (just starting) to 4 (excellent). Then your teacher scores it too. Talk about any row where the scores differ.

What we check	What excellent (4) looks like	Me	Teacher
Data	60 clearly labelled cards; labels checked; test cards kept apart.	1 2 3 4	1 2 3 4
Rules	A clear decision tree that uses features anyone can observe.	1 2 3 4	1 2 3 4
Testing	Accuracy worked out correctly on unseen cards; mistakes listed.	1 2 3 4	1 2 3 4
Improvement	A rule or the data is improved, and accuracy is measured again.	1 2 3 4	1 2 3 4
Fairness and honesty	Real results reported, and accuracy checked for each bin.	1 2 3 4	1 2 3 4
Talk and teamwork	A clear presentation; the work is shared fairly.	1 2 3 4	1 2 3 4

✓ Fair, safe and honest?

- [ ] We kept the test cards apart until the end.
- [ ] We reported our real accuracy, even when it was low.
- [ ] We checked unclear labels with our teacher instead of guessing.
- [ ] We asked which mistakes could cause harm, like a battery in the wet bin.

Looking back

Which items were hardest to classify, and why?

How could a wrong prediction cause harm in real life?

© 2026 Bullseye Knowledge System Limited

Answers

Many questions have more than one good answer. Accept any answer that follows the rule and can be explained.

1. Large Numbers Around Us

1. 100
2. b) 10 lakh
3. 7,73,300 and 3,00,377 (a number cannot start with 0)
4. 4,87,00,000
5. c) 0: anything multiplied by 0 is 0
Build it: 10,02,003 has 7 digits against 6, so it is bigger. 45,69,001 is bigger: the first difference is in the thousands place (7 against 9)
Challenge: 1,40,00,00,000 = 1,400,000,000, which is 1.4 billion

2. Arithmetic Expressions

1. 8 and 48
2. c) x (6 x 4 + 10 = 34)
3. 8 (23 - 5 + 8 - 6 + 12 = 32 passengers)
4. a) (2 + 3) x 4
5. 40: (3 + 5) x (9 - 4)
Build it: 3 x 45 + 2 x 120 - 50 = 135 + 240 - 50 = 325
Challenge: For example 4 x (5 + 3 - 2) = 24 or 2 x (3 + 4 + 5) = 24; accept any correct expression

3. A Peek Beyond the Point

1. b) 1101 (8 + 4 + 1)
2. 18 (16 + 2)
3. b) 6: 5 bits give only 32 numbers, 6 bits give 64
4. c) 0.405
5. a) 11.0 (alternately x 2 and + 0.2)
Build it: 14: remainder 0, n = 7; remainder 1, n = 3; remainder 1, n = 1; remainder 1, n = 0. Read backwards: 1110
Challenge: 10011 = 19 = S; R = 18 = 10010

4. Expressions using Letter-Numbers

1. 14 (20 - 6)
2. a) 5x + 4
3. c) 3 (B = 9, Z = 12)
4. 8: the middle row is 3 + P and P + 4, so 2P + 7 = 23
5. a) 14p (2p + 12p)
Build it: n = 10: 31 sticks. n = 17: 52 sticks, and it says 'Buy another box'
Challenge: 2n + 1; 51 sticks

5. Parallel and Intersecting Lines

1. 108°
2. x = 20; each angle is 70°
3. a) 3:00 and 9:00
4. a) parallel (the heading has turned 180°)
5. c) 30 (16 + 9 + 4 + 1)
Build it: An equilateral triangle. Rhombus: repeat 2 times: forward 60, turn right 60°, forward 60, turn right 120°
Challenge: 4 angles are 50° and the other 4 are 130°

6. Number Play

1. 16
2. b) 4
3. a) all ON (UUU)
4. c) 9, because (10a + b) - (10b + a) = 9(a - b)
5. b) no: two odd digits make an even sum, and even digits keep it even
Build it: Only 74: 9 x (a - b) = 27 means a - b = 3, and a + b = 11, so a = 7 and b = 4 (74 - 47 = 27)
Challenge: 32 cards; 3 are left (Y on, P off, and two of R, G and B on)

7. A Tale of Three Intersecting Lines

1. c) 5, 7, 10
2. 65°
3. b) right-angled
4. b) 2: 6, 6, 8 and 8, 8, 4 (4, 4, 12 and 2, 2, 16 break the triangle inequality)
5. a) the square (16 cm^2, the triangle is about 6.9 cm^2)
Build it: It says 'yes' (2 + 10 > 3), but 2 + 3 < 10, so it is not a triangle. Fix: If a + b > c AND b + c > a AND a + c > b
Challenge: 3: (2, 5, 5), (3, 4, 5) and (4, 4, 4)

8. Working with Fractions

1. 36
2. a) 3/5 (18/30)
3. 80: before B there were 80; before the 20 got on, 60; that is 3/4 of the start
4. a) 8
5. 1/5
Build it: 600, then 600 - 200 = 400, then 400 - 100 = 300: it says 300
Challenge: Set money to 300; divide money by 3/4 (400); divide money by 2/3 (600). Undo the last step first

AI 1. AI Domains and Applications

1. b) regression
2. c) clustering
3. b) Computer Vision
4. 75
5. c) NLP
Build it: Two groups: small, very sweet fruit (like grapes or berries) and big, less sweet fruit (like apples or guavas). Clustering finds them without labels
Challenge: Open answer, for example: sorting library books by subject (classification); predicting lunch numbers (regression); grouping students by favourite clubs (clustering)

© 2026 Bullseye Knowledge System Limited

AI 2. AI in Industries

1. a) healthcare
2. b) education
3. c) transport
4. c) C (10 km), even though it is the slowest
5. a) AI can make mistakes, and the doctor is responsible for the patient
Build it: Road 1 gets a longer green light; when the ambulance is detected, road 2 turns green at once
Challenge: The ambulance check, because an emergency must win over any queue; checked last, the car rule could delay it

AI 3. Data Visualisation and Analysis

1. b) line graph
2. a) pie chart
3. 90° (15/60 x 360)
4. a) precise but not accurate
5. a) a table of marks
Build it: A line graph. Sales rise through the week and are highest at the weekend (40 on Saturday)
Challenge: The bar heights no longer show the real values; start the axis at 0, or clearly mark the break in the axis

AI 4. Ethics and AI Bias Awareness

1. b) historical bias
2. a) data bias
3. c) measurement bias: spending is not the same as illness
4. a) a person should review it, because important decisions need a human in the loop
5. a) check it, and never share posts that hurt someone
Build it: Fruit 92%, vegetable 90%, overall 55 of 60 (about 92%). Yes: both groups are now about equally accurate
Challenge: Open answer, for example: a teacher makes the final decision; check that the AI works equally well for every group; explain to students how it was used

Project 1. Nutri-Quest Food Planner

Open project: assess with the rubric. Check that every day of the final plan passes every rule.

Project 2. Sort It with AI

Open project: assess with the rubric. Accuracy = correct test cards ÷ all test cards, for example 17 ÷ 20 = 85%.

© 2026 Bullseye Knowledge System Limited

Coder words

accuracy	closeness to the true value	AI 3
balanced	has every food group	Project 1
bias	an unfair preference	AI 4
binary	place values in twos	Ch 3
bit	a binary digit, 0 or 1	Ch 3
brackets	do this part first	Ch 2
budget	the most you can spend	Project 1
case	one possible combination	Ch 6
classification	sort into labelled groups	AI 1
clustering	find groups without labels	AI 1
constraint	a rule a plan must follow	Project 1
crore	1,00,00,000	Ch 1
dataset	the data a model learns from	AI 1
decimal	place values in tens	Ch 3
diagnosis	finding out what illness it is	AI 2
equilateral	all sides equal	Ch 7
estimate	a close, easy answer	Ch 1
expression	numbers joined by operations	Ch 2
fairness	works equally well for everyone	AI 4
feature	something you can observe	Project 2
filter	remove cases that break a rule	Ch 6
formula	a rule written with letters	Ch 4
human in the loop	a person checks the decision	AI 4
intersect	cross at a point	Ch 5
inverse	the step that undoes another	Ch 8
isosceles	two sides equal	Ch 7
label	the right answer for an example	Project 2
lakh	1,00,000	Ch 1
like terms	terms with the same letter	Ch 4
line graph	shows change over time	AI 3
million	10,00,000 (10 lakh)	Ch 1
nutrient	what food gives the body	Project 1
of	means multiply	Ch 8
parallel	never meet	Ch 5
personalised	made for one person	AI 2
pie chart	shows parts of a whole	AI 3
place value	what a digit's position is worth	Ch 3

© 2026 Bullseye Knowledge System Limited

precedence	which operation goes first	Ch 2
reciprocal	the fraction turned upside down	Ch 8
regression	predict a number	AI 1
route planning	choosing the best way	AI 2
scalene	no sides equal	Ch 7
structured data	data in rows and columns	AI 3
substitute	put a number in place of a letter	Ch 4
term	a part added or subtracted	Ch 2
test set	examples kept back for checking	Project 2
translation	changing one language to another	AI 2
transparency	being open about how it decides	AI 4
transversal	a line crossing two lines	Ch 5
triangle inequality	each side < sum of the other two	Ch 7
true or false	the two values in logic	Ch 6
truth table	a list of every case	Ch 6
variable	a letter that stands for a number	Ch 4
vertically opposite	equal angles across a crossing	Ch 5
work backwards	start from the end	Ch 8

© 2026 Bullseye Knowledge System Limited

Cut-out code cards

Ask an adult to help you cut along the dotted lines. Use the cards to build programs on your desk, then act them out with a friend.

Define ___	give back ___	Repeat until ___
Define ___	give back ___	Repeat until ___
Repeat for each ___	If ___	else
Repeat for each ___	If ___	else
AND	OR	Input ___
AND	OR	Input ___
Set ___ to ___	change ___ by ___	say ___
Set ___ to ___	change ___ by ___	say ___

© 2026 Bullseye Knowledge System Limited